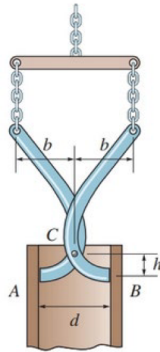


Take-Home Assignment weeks 37–38

Noah Rahbek Bigum Hansen

Exercise P5-1

The pipe is hoisted using the tongs. If the coefficient of static friction at A and B is μ_s , determine the smallest dimension b so that any pipe of inner diameter d can be lifted.



The normal conditions for equilibrium states:

$$F_{R_x} = 0$$

$$F_{R_y} = 0$$

$$M_{R_C} = 0.$$

At each contact point, A and B , the impending slip is downward on the tong, so the resultant

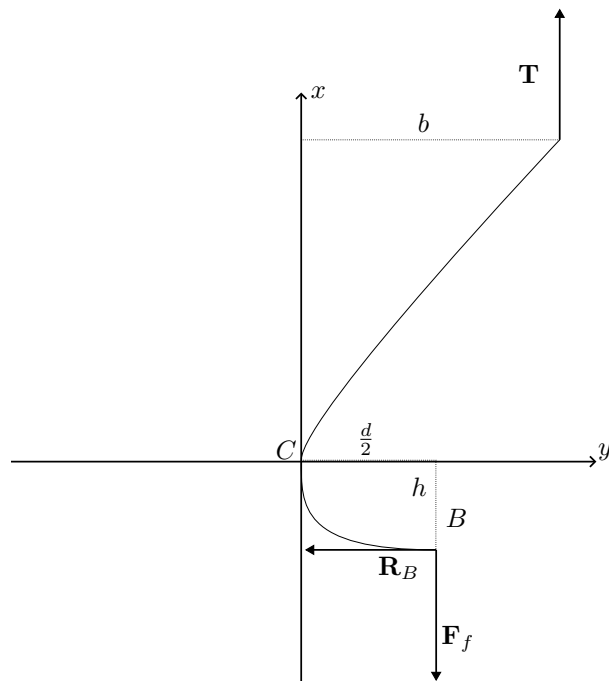


Figure 0.1: Free-body diagram